

Factors Contributing to Traumatic Brain and Spinal Cord Injury in Massachusetts Motor Vehicle Crashes

Massachusetts Crash-Related Injury Surveillance System, 2012 - 2015



Injury Surveillance Program, Massachusetts Department of Public Health

October 2021



Massachusetts Crash-Related Injury Surveillance System

The Massachusetts Crash-Related Injury Surveillance System (MA CRISS) includes data for persons treated in MA acute care hospitals for motor vehicle crash injuries whose hospital record linked with a MA police crash report. These data do not include all crashes involving injuries in MA, as they do not include cases in which crash victims were transported to out-of-state hospitals, police were not involved, crash reports were not submitted to the Registry of Motor Vehicles, or missing or incorrect data prevented data linkage. Data may contain some duplicate records and/or linkages of some hospital records with the wrong crash records.

MA Hospital Discharge data are compiled by the Center for Health Information and Analysis. Crash data are compiled by the MA Registry of Motor Vehicles.

Acknowledgements: Data linkage and analyses were conducted by Catherine Rahilly-Tierney, MD, MPH and Arman Altincatal, MS, of Strategic Research Partners, LLC, on behalf of the MA Department of Public Health, Injury Surveillance Program.

This report explores the circumstances surrounding motor vehicle crashes that result in traumatic brain and spinal cord injuries. Understanding those circumstances can inform prevention programs and policies so fewer people suffer these serious injuries and potential long-term disability.¹

After falls, motor vehicle crashes are the second leading cause of death and hospitalization involving traumatic brain injury (TBI) in Massachusetts (MA).² In 2015, of the 197 car/truck drivers^a killed in a crash in MA, approximately two out of five (41%) sustained a TBI in the crash.^{3,b}

Of the 1,219 hospitalizations of drivers in MA for crash-related injuries in fiscal year (FY) 2015, 30% sustained a TBI and 3% sustained a spinal cord injury (SCI).⁴ DPH found no recent estimates of TBI or SCI (TBI/SCI) frequency among hospitalized drivers in other states, but a study of drivers in Maryland hospitalized for crash-related injuries between 1994-1996 found that 38% sustained a TBI in the crash.⁵

Based on FY 2015 data, hospitalized drivers who sustained a TBI/SCI in the crash had longer hospital stays (7 vs. 5 days) and higher mean

In 2015, over 40% of drivers killed and 30% of drivers hospitalized in Massachusetts sustained a traumatic brain injury in the crash.

Prior studies have identified not wearing a seatbelt, lateral impact, airbag non-deployment, and age under 40 as risk factors for sustaining a TBI in a crash.^{5,6,7} There are mixed results as to whether males or females are at higher risk of sustaining a TBI in a crash.^{5,6}

This study used linked 2012 - 2015 MA Crash-Related Injury Surveillance System (MA CRISS) data to identify driver and crash-related factors associated with sustaining a TBI/SCI in a crash. Note that MA CRISS data differ from the hospitalization data presented above, in that MA CRISS data include only those hospital records that link to a police crash report. See box on left for further details.

This study focused on drivers only; passengers were excluded. A separate report examines factors associated with hospitalized motorcycle operators sustaining a TBI/SCI in a crash. We combined TBI and SCI because they are both neurological injuries associated with significant long-term disability.

^a In this report, "drivers" refers to drivers of cars, vans, or trucks. Motorcycle operators are covered in a separate report.

^b Spinal cord injury diagnoses are under-reported in death data, so were not included here.

Characteristics of Drivers and Related Crashes in MA CRISS Data

Of 4,060 hospitalized drivers in 2012-2015 MA CRISS data (Jan. 1, 2012 - Sep. 30, 2015), nearly one in three (30%) sustained a TBI or SCI in the crash (TBI - 29% and SCI - 2%).^c These percentages are similar to those described in FY 2015 hospitalization data above. TBI/SCI and demographic data were obtained from hospital discharge data.^d Seatbelt use, speeding, lane departure^e, airbag deployment, and point-of-impact data were obtained from crash data. Data on alcohol/drug intoxication were obtained from both of these data sources. Table 1 describes these characteristics in hospitalized drivers who sustained a TBI/SCI in the crash and those who did not sustain a TBI/SCI in the crash.

When looked at individually, most factors studied were significantly associated with sustaining a TBI/SCI in the crash. Specifically, of hospitalized drivers who sustained a TBI/SCI the crash:

- 41% were identified as not wearing a seatbelt compared with 24% of those who did not sustain a TBI/SCI.ⁱ
- 17% were identified as speeding compared with 11% of those who did not sustain a TBI/SCI.
- 29% were identified as intoxicated on alcohol or drugs compared with 23% of those who did not sustain a TBI/SCI.
- 30% had documentation that airbags did not deploy in the crash compared with 25% of those who did not sustain a TBI/SCI.^k
- A higher percentage of drivers who were male, ages 46 and under, and people of color



Table 1. Driver and Crash-related Factors in Hospitalized Drivers who Sustained and did not Sustain a TBI/SCI in the Crash.

<u>Factor</u>	<u>Sustained a TBI/SCI n (%)^f</u>	<u>Did not sustain a TBI/SCI n (%)^f</u>
<u>Total</u>	<u>1,213 (100%)</u>	<u>2,847 (100%)</u>
<u>Ages^g*</u>		
<u>Ages 46 and under</u>	<u>688 (57%)</u>	<u>1,306 (46%)</u>
<u>Ages 47 and older</u>	<u>525 (43%)</u>	<u>1,541 (54%)</u>
<u>Sex^g*</u>	-	-
<u>Male</u>	<u>765 (63%)</u>	<u>1,577 (55%)</u>
<u>Female</u>	<u>448 (37%)</u>	<u>1,270 (45%)</u>
<u>Race/ethnicity^h*</u>	-	-
<u>White, non-Hispanic</u>	<u>943 (81%)</u>	<u>2,359 (86%)</u>
<u>People of color</u>	<u>217 (19%)</u>	<u>377 (14%)</u>
<u>Alcohol/drug intoxication[*]</u>	-	-
<u>Yes</u>	<u>354 (29%)</u>	<u>665 (23%)</u>
<u>Not documented</u>	<u>859 (71%)</u>	<u>2,182 (77%)</u>
<u>Seatbelt useⁱ*</u>	-	-
<u>Seatbelt worn</u>	<u>505 (59%)</u>	<u>1,561 (76%)</u>
<u>Seatbelt not worn</u>	<u>352 (41%)</u>	<u>500 (24%)</u>
<u>Speeding[*]</u>	-	-
<u>Yes</u>	<u>206 (17%)</u>	<u>327 (11%)</u>
<u>Not documented</u>	<u>1,007 (83%)</u>	<u>2,520 (89%)</u>
<u>Lane departure[*]</u>	-	-
<u>Yes</u>	<u>785 (65%)</u>	<u>1,677 (59%)</u>
<u>Not documented</u>	<u>428 (35%)</u>	<u>1,170 (41%)</u>
<u>Point-of-impact^j</u>	-	-
<u>Lateral impact</u>	<u>301 (25%)</u>	<u>711 (25%)</u>
<u>Front/rear impact</u>	<u>904 (75%)</u>	<u>2,104 (75%)</u>
<u>Airbag deployment^k*</u>		
<u>Airbag deployed</u>	<u>758 (70%)</u>	<u>1,921 (75%)</u>
<u>Airbag not deployed</u>	<u>329 (30%)</u>	<u>636 (25%)</u>

^c 22 drivers sustained both a TBI and a SCI.

^d ICD-9-CM codes in any field were used to identify TBI (800-801.

803-804, 850-854.19, 950(.1-.3) or 959.01) and SCI (806 or 952).

^e Non-intersection crash in which a vehicle crosses an edge line, a centerline, or otherwise leaves the traveled way.

^f Denominator changes based on missing data.

^g Age was divided into two groups at the median age of 47.

^h Excludes 164 cases where race/ethnicity was missing.

ⁱ Excludes 1,142 cases where seatbelt use was missing.

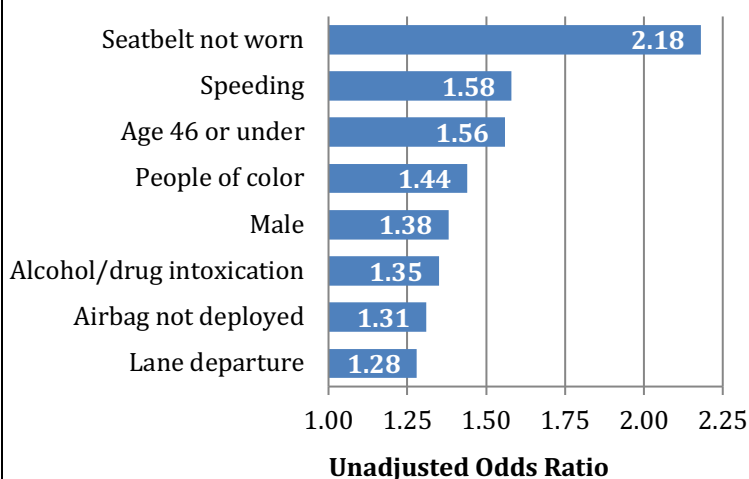
^j Excludes 40 cases where point-of-impact was missing.

^k Excludes 416 cases where airbag deployment status was

Driver and Crash-related Factors Associated with Sustaining a TBI/SCI in a Crash

We calculated unadjusted odds ratios (OR)^k to further understand factors that were significantly associated with sustaining a TBI/SCI in a crash. Figure 1 shows only the factors with ORs that were significantly associated with sustaining a TBI/SCI in a crash ($p < 0.05$).

Figure 1. Odds Ratios for Factors Significantly Associated with Sustaining a TBI/SCI in a Crash, Hospitalized Drivers, 2012-2015 MA CRISS data (N =



In this sample of hospitalized drivers:

- Not wearing a seatbelt more than doubled the odds (OR = 2.18) of drivers sustaining a TBI/SCI in the crash, compared to drivers identified as wearing a seatbelt.
- Speeding increased the odds of sustaining a TBI/SCI in the crash by 58% (OR = 1.58), compared to drivers not identified as speeding.
- Alcohol/drug intoxication increased the odds of sustaining a TBI/SCI in the crash by 35% (OR = 1.35), compared to drivers not identified as intoxicated.
- Drivers who were age 46 or under, a person of color, or male had increased odds of sustaining a TBI/SCI in the crash of 56%, 44%, and 38% respectively (ORs = 1.56, 1.44, and 1.38), compared to drivers who were age 47 or older, White non-Hispanic, or female.
- Airbag non-deployment increased the odds of sustaining a TBI/SCI in the crash by 31% (OR =

^k Unadjusted odds ratios do not take other potential contributing factors into account.

¹ Lt. Timothy Dowd, personal communication, June 24, 2019.



In addition to these associations with sustaining a TBI/SCI, many factors were associated with one another. Among drivers who sustained a TBI/SCI, factors significantly associated with one another included: alcohol/drug intoxication, speeding, not wearing a seatbelt, and lane departures. Being male or age 46 or under were also significantly associated with these four risk factors in drivers who sustained a TBI/SCI, although being a person of color was not associated with these four factors (data not shown).

The increased risk of TBI/SCI in people of color may be related to vehicle characteristics that were not included in this study. Prior studies have found that Hispanic motorists and communities with lower income are more likely to have older vehicles with fewer safety features than non-Hispanic, white motorists or communities with higher income.¹⁰

Airbag deployment was significantly associated with driver's age. Of crashes in which drivers sustained a TBI/SCI, airbags were not deployed in nearly half of crashes involving drivers ages 75 and older, compared to one-quarter of crashes involving drivers under age 55 (data not shown). Prior studies have found that older drivers are more likely to be involved in low-speed crashes¹⁰, which may be one reason for airbag non-deployment.¹

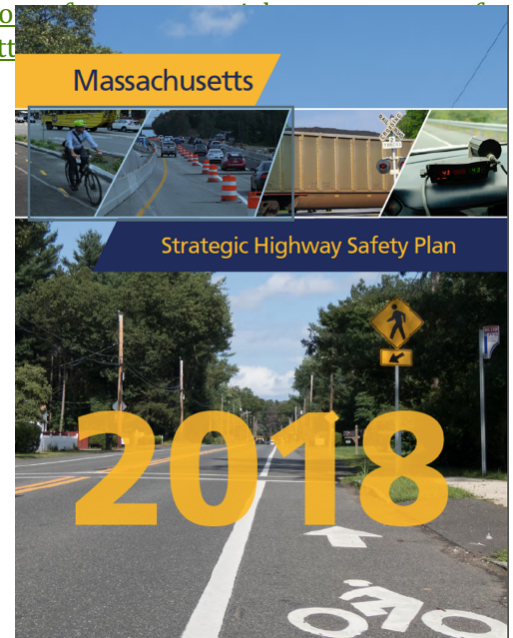
Limitations: TBI cases may be underestimated, as milder TBI cases may not get diagnosed until after patients are discharged from the hospital. Intoxication rates may also be underestimated, as healthcare providers and police may not always test drivers for alcohol/drugs, or test results may not be documented in the hospital or crash record. Missing crash data may limit the accuracy of several factors, including seatbelt use and airbag deployment. Speeding and driving under the influence of alcohol/drugs may also be underestimated due to incomplete violation codes

Strategies to Prevent Traumatic Brain and Spinal Cord Injuries in Motor Vehicle Crashes

Preventing TBI and SCI that result from motor vehicle crashes requires increasing seatbelt use, preventing impaired driving, and preventing speeding crashes. This analysis/study identified that in Massachusetts, men, people of color, and drivers under 47 years of age are at higher risk of sustaining these injuries, and therefore prevention strategies must be developed or evaluated with those populations in mind. In addition, prevention strategies and policies should address and account for racial bias and inequity. The following describes ways Massachusetts

Support Primary Seatbelt Legislation

Proposed in the MA Strategic Highway Safety Plan (SHSP) 2018, primary seatbelt legislation is the most effective strategy for increasing seatbelt usage. Massachusetts is one of 15 states that have a secondary seatbelt law, meaning that a police officer cannot stop and ticket a driver for the sole offense of not wearing a seatbelt. Primary Seatbelt Legislation can increase seatbelt use by 8%, and seatbelt use during a crash is known to reduce fatalities by 45%.¹¹ In states that strengthened laws from secondary to primary enforcement, driver death rates declined by 7% annually over a 15-year period (1989-2003).¹² Implementation of this evidence-based approach should integrate considerations around equitable enforcement of such a law. Policymakers should engage communities of color throughout the process of developing, piloting, and evaluating primary seatbelt legislation. Furthermore, primary seatbelt legislation should provide for the collection of race and ethnicity data for cited drivers.



Impaired Driving Prevention

Proposed in the MA Strategic Highway Safety Plan (SHSP) 2018, All-Offender Ignition Interlock legislation would permit judges to require ignition interlock devices for first time OUI offenders. Studies show that ignition interlock devices can reduce re-arrest rates for impaired driving by as much as 70% and reduce alcohol-involved crash deaths by 15%.¹¹ All-offender Ignition Interlock Legislation should be coupled with the establishment of a fund for individuals who cannot afford the ignition interlock installation and maintenance fees, similar to the fund for Section 24D driver alcohol education program. First time offender ignition interlock legislation should include provisions for collection of demographic data of drivers sentenced under those policies.

Additionally, programs related to driving under the influence of alcohol programs are operated by the MA Department of Public Health Bureau of Substance Addiction Services. These programs serve as alternatives to loss of license or incarceration:

- First Offender Driver Alcohol Education (DAE): 40-hour educational program to help drivers identify and understand alcohol addiction and drinking-and-driving behaviors.
- Second Offender Driving Under the Influence of Liquor (DUIL) Program: 14-day residential program that includes medical evaluation, individual and group counseling, educational sessions, and assignment to a Second Offender Aftercare Program.
- Second Offender Aftercare (SOA) Program: one-year program of individual, group, and family services individually designed to meet the needs of the client.

Improving access to alternative transportation programs is another strategy for preventing impaired driving. For more information about alternative transportation programs, see Alternative Transportation Programs: A Countermeasure for Reducing Impaired Driving.

Additional strategies to prevent impaired driving are enumerated in the Federal Fiscal Year 2020 MA Highway Safety Plan.

Strategies to Prevent Traumatic Brain and Spinal Cord Injuries in Motor Vehicle Crashes (cont.)

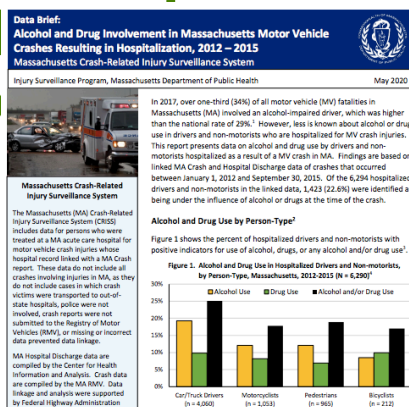
Culturally and Linguistically Appropriate Education

It must become common knowledge that not using a seatbelt, driving while intoxicated, or speeding may not only end in fatality, but can also result in life-altering disabilities such as paralysis and brain injury. Community-level, culturally and linguistically appropriate outreach and education geared towards populations at higher risk for sustaining such injuries is necessary to share that message, especially if the above legislative changes are implemented. Outreach and education materials should be developed with input (e.g., collecting qualitative data through focus groups and key informant interviews) from high-risk populations and delivered through trusted sources such as community health workers, community leaders, employers, community support specialists, and others.

National School Reform Agreement:

2020 Annual Report on progress

Preventing traumatic brain and spinal cord injuries that result from motor vehicle crashes requires a combination of driving behaviors and circumstances that contribute to crash risk and evaluation of strategies to reduce crash risk. Efforts to support, sustain, analyze, and disseminate data from MA CRISS will continue to improve our ability to prevent crash-related injuries and deaths.



Data Sources and References

Data Sources in the MA Crash-Related Injury Surveillance System used in these analyses:

- Inpatient Hospital Discharge data (Jan. 2012 – Sep. 2015), Center for Health Information and Analysis
- Crash Data System (Jan. 2012 – Sep. 2015), MA Registry of Motor Vehicles

- ¹ Chia-Ying Kuo, et al. Functioning and Disability Analysis of Patients with Traumatic Brain Injury and Spinal Cord Injury by Using the World Health Organization Disability Assessment Schedule 2.0. *Int J Environ Res Public Health*. 2015 Apr 14;12(4):4116-27.
- ² Massachusetts Department of Public Health, Injury Surveillance Program (June 2018). *Massachusetts Special Emphasis Report: Traumatic Brain Injury – 2015*. Retrieved from <https://archives.lib.state.ma.us/handle/2452/802168>
- ³ MA Department of Public Health. *Registry of Vital Records and Statistics*.
- ⁴ Center for Health Information and Analysis. MA Inpatient Hospital Discharge data. Fiscal year Oct. 1, 2014 – Sep. 31, 2015.
- ⁵ Dischinger PC, Ho SM, & Kufera JA (1999). The Epidemiology of Traumatic Brain Injury: A Statewide Study of Hospitalized Maryland Drivers. *Annu Proc Assoc Adv Automot Med*. 1999; 43: 71–88.
- ⁶ Antona-Makoshi J, et al. Accident analysis to support the development of strategies for the prevention of brain injuries in car crashes. *Accid Anal Prev*. 2019; 117:98-105.
- ⁷ Siegel JH, et al. Factors influencing the patterns of injuries and outcomes in car versus car crashes compared to sport utility, van, or pick-up truck versus car crashes: Crash Injury Research Engineering Network Study. *J Trauma*. 2001 Nov;51(5):975-90.
- ⁸ Harper JS, et al. Motor vehicle crash fatalities: a comparison of Hispanic and non-Hispanic motorists in Colorado. *Ann Emerg Med*. 2000;36(6):589-596.
- ⁹ Girasek DC, Taylor B. An exploratory study of the relationship between socioeconomic status and motor vehicle safety features. *Traffic Inj Prev*. 2010;11(2):151-155.
- ¹⁰ Kent R, et al. On the fatal crash experience of older drivers. *Annu Proc Assoc Adv Automot Med*. 2005;49:371-91.
- ¹¹ Massachusetts Department of Transportation (2018). *Massachusetts Strategic Highway Safety Plan*. Retrieved from <https://www.mass.gov/doc/massachusetts-shsp-2018/download>
- ¹² Farmer CM, Williams AF. Effect on fatality risk of changing from secondary to primary seat belt enforcement. *J Safety Res*. 2005; 36(2): 189-94

This study and report were supported by MA Department of Transportation (MassDOT) funds and Centers for Disease Control and Prevention (CDC) grant #NU17CE924835-01-00. The content is solely the responsibility of the authors and does not necessarily represent the official views of MassDOT or the CDC.

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National School Reform Agreement: 2020 Annual Report on progress in implementing the national policy initiatives

This report outlines the progress towards implementing the national policy initiatives in 2020 against agreed milestones contained in the *National School Reform Agreement*. It is the second annual public report, as required by clause 53.a of the Agreement and follows the publication of the 2019 progress report in September 2020.

The Agreement features three reform directions that focus on supporting students, student learning and achievement; supporting teaching, school leadership and school improvement; and enhancing the national evidence base.

The reform directions are being progressed nationally through the eight national policy initiatives, together with a suite of state-specific activities outlined in bilateral agreements between the Commonwealth and states or territories.

The national policy initiatives are being developed in close consultation and collaboration with government and non-government education authorities and broader education stakeholders including, but not limited to, teachers, school leaders and parents.

The 2019-2020 bushfire season and the COVID-19 pandemic presented significant challenges for the implementation of the national policy initiatives in 2020 and changed the way we consult and collaborate. Despite these challenges, governments progressed the national policy initiatives with minor delays.

In June 2020 Education Council agreed minor adjustments to the Agreement's Schedule B milestones. These were formalised by Education Council on 4 September 2020 and a revised Agreement published on 2 October 2020. This report notes progress against the revised milestones.

The 2020 report shows that there has been good progress across all of the eight national policy initiatives, with two notable achievements:

- The review of senior secondary pathways was completed as scheduled in June 2020 and the report, *Looking to the Future: Report of the Review of senior secondary pathways into work, further education and training*, was published in July 2020. In December 2020, Education Council agreed to focus on areas that could be actioned immediately and address potential gaps in current activity, including literacy, numeracy and digital literacy; developing a sample Learner Profile; and agreeing a common language for skills and capabilities. In 2021, education and skills ministers will work together to progress this reform agenda, including improved career guidance through the National Careers Institute and development of a national strategy on VET delivery to secondary school students.
- In December 2020, the national evidence institute was launched as the Australian Education Research Organisation (AERO). AERO has been established to drive improvements in student outcomes through improving teachers' access to evidence about what works in the classroom. AERO delivers against a key recommendation of the *Through Growth to Achievement Report of the Review to Achieve Educational Excellence in Australian Schools* (2018).

On 26 June 2020, National Cabinet agreed that former Director General of the Western Australian Department of Premier and Cabinet and former Commonwealth Cabinet Secretary Mr Peter Conran AM would lead a review of the former COAG Councils and Ministerial Forums with a view to rationalise and reset their work.

On 23 October 2020, National Cabinet accepted the Review's recommendations that include reducing the number of ministerial forums and ensuring that those that remain are more agile and responsive, with direct responsibility for decision making resting with Ministers. Consistent with the direction set by National Cabinet, ministers' meetings will not report to National Cabinet (or the National Federation Reform Council) unless tasked.

This represents a significant reform to Commonwealth-State relations and will significantly reduce bureaucracy and red tape. National Cabinet also noted that the range of significant reforms agreed have come about directly through the collaboration of National Cabinet, as the country comes together to continue to respond to the COVID-19 pandemic.

As part of these reforms, National Cabinet agreed to maintain and reset Education Council as the Education Ministers Meeting.

From 2021, all Australian governments will continue to collaborate on strategies to address the long-term challenges facing Australia's education systems through the new Education Ministers. The reset forum for Ministers provides the opportunity for a renewed focus on key strategic priorities and an emphasis on delivering effective outcomes under the Agreement.

Under clauses 51–55 of the *National School Reform Agreement*, all parties have committed to reporting on progress in implementing agreed national policy initiatives outlined in Schedule B to give the community confidence that Education Ministers are making efforts to improve the quality and equity of Australia's schooling system. Table 1 of this report reflects progress as at 31 December 2020 against revised *National School Reform Agreement* Schedule B milestones approved by Education Ministers on 4 September 2020.

Table 1: Progress in the implementation of the national policy initiatives against milestones in Schedule B of the *National School Reform Agreement*

National Policy Initiative	Milestone	Timing	Has the milestone been met? Yes, partially, or not yet applicable
A (i) Enhancing the Australian Curriculum to support teacher assessment of student attainment and growth against clear descriptors	a) Education Council consider proposal(s) for the development of learning progressions across the Australian Curriculum learning areas and general capabilities, including consideration of existing progressions and development stages, timelines and estimated costs.	Early 2019	Yes. More information is available from the Online Formative Assessment Initiative website .
	b) Subject to Education Council agreement at a), develop and trial learning progressions and formative assessment capability for learning areas and general capabilities, with priority on literacy and numeracy in the early years of schooling.	Early 2021	Not yet applicable.
	c) Subject to Education Council agreement to a) and b) above, existing and new learning progressions and formative assessment capability for agreed Australian Curriculum learning areas and general capabilities are progressively made available.	From 2022	Not yet applicable.
A (ii) Assisting teachers monitor individual student progress and identify student learning needs through opt-in online and on-demand student learning assessment tools with links to student learning resources, prioritising early years foundation skills	a) Education Council consider a proposal for the enhancement of formative assessment capabilities, including consideration of existing IT solutions, online assessment resources and digital learning resources and potential benefits to students, teachers and school leaders. It is expected the proposal will include development stages, timelines, governance arrangements and estimated costs.	Early 2019	Yes. More information is available from the Online Formative Assessment Initiative website .
	b) Subject to Education Council agreement at a), Education Council commission: - the design, build and piloting of an opt-in formative assessment facility with priority on early years literacy and numeracy. Assessments should be matched to agreed learning progressions (such as English, mathematics, science and critical and creative thinking), and - the development of supporting professional learning modules to build teacher and school leader capacity and capability to use effective formative assessment practices.	Early 2021	Not yet applicable.

National Policy Initiative	Milestone	Timing	Has the milestone been met? Yes, partially, or not yet applicable
	e) — Subject to Education Council agreement at a) and b), progressively develop and release nationally available online on-demand formative assessment resources (linked to the development schedule for learning progressions across Australian Curriculum learning areas and general capabilities), refine the online formative assessment facility based on feedback from the pilot and align digital teaching resources to the learning progressions.	From 2022	Not yet applicable.
A (iii) Reviewing senior secondary pathways into work, further education and training	a) — Education Council agree panel membership and terms of reference for a review of senior secondary education, pathways to work, further education and training and consideration of prerequisites for university entry.	Early 2019	Yes. More information is available on the Department of Education Skills and Employment website
	b) — Subject to Education Council agreement at a), Education Council receive review report and consider recommendations for any national initiatives to strengthen student pathways, with States and Territories to independently consider recommendations for local response.	June 2020	Yes.
	c) — Subject to Education Council agreement at a) and b), Education Council has implemented, or is progressing, agreed recommendations as appropriate.	End of the agreement	Yes.
B (i) Reviewing teacher workforce needs of the future to attract and retain the best and brightest to the teaching profession and attract teachers to areas of need	a) — Education Council consider an implementation strategy for the recommendations of the National Review of Teacher Registration.	Early 2019	Yes.
	b) — Education Council begin to develop options for a national teacher workforce strategy.	Mid-2019	Yes. More information is available from the Australian Institute for Teaching and School Leadership .
	c) — Subject to b), Education Council agree to an option for a national teacher workforce strategy to support decision making of teacher employers and initial teacher education (ITE) providers. It is anticipated the strategy would identify requirements to address workforce issues of supply, demand, retention, attrition, hard to staff schools, specialisation, teaching out of field, rural and remote workforce and any gaps in data identified by Parties and sectors.	September 2020	Yes, through Education Ministers' 11 December 2020 endorsement of the <i>National Initiatives to Support Teaching and School leadership narrative</i> , available on the Department of Education Skills and Employment website .

National Policy Initiative	Milestone	Timing	Has the milestone been met? Yes, partially, or not yet applicable
	d) — Subject to Education Council agreement at b) and e), Education Council implement a national teacher workforce strategy to respond to teacher workforce needs of the future and support decision making of teacher employers and ITE providers.	From December 2020	Yes. The Australian Institute for Teaching and School Leadership reported to Education Ministers in December 2020 with the report <i>Teaching Futures: A National Teacher Workforce Strategy for Australia</i> . This paper has helped shape the future direction of national and collaborative work on teaching. This paper and the <i>National Initiatives to Support Teaching and School Leadership</i> narrative meet the commitment to ‘review teacher workforce needs’.
B (ii) Strengthening the initial teacher accreditation system	a) — States and Territories ensure that accredited initial teacher education programs require pre-service teachers to have successfully completed an endorsed final year teaching performance assessment (TPA) prior to graduation.	From January 2019	Partially.
	b) — Commence national quality assurance activities in cooperation with all jurisdictional authorities.	From January 2019	Yes.
C (i) Implementing a national unique student identifier (USI) that meets national privacy requirements in order to support better understanding of student progression and improve the national evidence base	a) — Education Council consider: i. — a proposal for a national USI to enhance school and system self review and external quality assurance, including proposed approach and compatibility with existing USIs ii. — Governance, privacy protections, national data access protocols and protections and estimates and proposed allocations of costs.	Late 2019	Yes.
		September 2020	Partially.
	b) — Subject to Education Council agreement at a), implement the institutional arrangements necessary to support the implementation of a national USI.	March 2021	Not yet applicable.

National Policy Initiative	Milestone	Timing	Has the milestone been met? Yes, partially, or not yet applicable
	c) Subject to Education Council agreement at a) and b), and subject to meeting national privacy requirements, schools and systems to progressively work to create a national USI for each student from 2021.	All students to have a USI by the end of the agreement	Not yet applicable.
C (ii) Establishing an independent national evidence institute to inform teacher practice, system improvement and policy development	a) Education Council consider options for an independent national evidence institute, including scope of functions, potential for expansion of existing capabilities, interaction with existing national bodies, funding and governance arrangements.	Late 2019	Yes.
	b) Subject to Education Council agreement to an option at a), implement the institutional arrangements necessary to support an independent national evidence institute.	Mid 2020	Yes. More information is available from the Australian Education Resource Organisation website .
	c) Subject to Education Council agreement at a) and b), the national evidence institute commences operations from 2020, and inter alia, develops and implements a national research and evidence plan.	From 2020	Yes.
C (iii) Improving national data quality, consistency and collection to improve the national evidence base and inform policy development.	a) Education Council agree equity and proficiency standards (for numeracy and literacy assessed by NAPLAN) as part of the scheduled review of the <i>National Measurement Framework for Schooling</i> .	December 2018	Yes.
	b) Education Council consider strategic opportunities to enhance the national evidence base, including options for measures of child development status at school entry, student learning gain, general capabilities, post school destination information, attainment, retention, wellbeing, post school outcomes.	Life of the agreement	Partially.